

## 3004ENG — Project Management Principles

Trimester 2, 2026

### Eris TestFlight2 Launch Campaign

#### Project Execution Plan

Assessment 3: Group Project Execution Plan — Group 1

|                        |                                             |
|------------------------|---------------------------------------------|
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Acting as a project management consultancy engaged to deliver launch-site operations at Bowen Orbital Spaceport.

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# **1 Executive summary**

## **2 Refined project overview**

### **2.1 Background and purpose**

### **2.2 Scope**

### **2.3 Charter baseline versus PEP refinement**

### **2.4 Objectives and success criteria**

### **2.5 Work breakdown structure**

The work breakdown structure in Figure 2.1 is the 100% rule applied to launch-site operations at Bowen Orbital Spaceport (Project Management Institute, 2021). Four Level-2 workstreams—campaign governance, pad logistics, pad testing, and flight/closeout—decompose into seventeen Level-3 packages. Every scheduled activity except A-100 maps to exactly one of those packages, and every dollar in Table 4.1 sits in exactly one of them. A-100 is annotated beside the tree rather than inside it: the vehicle receipt window is an external zero-dollar wait that starts when out-of-scope freight reaches the gate. It belongs on the network because it drives stacking (A-122), but it is not a cost account.

Package 1.2.1 is one cost account with two schedule bars (A-121 site mobilisation and A-122 receiving inspection) because receiving cannot start until both the cleanroom and the vehicle exist. The remaining fifteen Level-3 codes are one-to-one with Table 3.1. Factory manufacture, engine research and development, and line-haul to the receiving gate are excluded; parking them as unidentified Level-3 boxes would violate the 100% rule.

Limited notice-to-proceed in the 7 August charter authorised field mobilisation from 10 August. The WBS is therefore live before Gate 1. What Gate 1 (A-111, 2 October) locks is the cost and schedule baseline, not the right to occupy the pad.

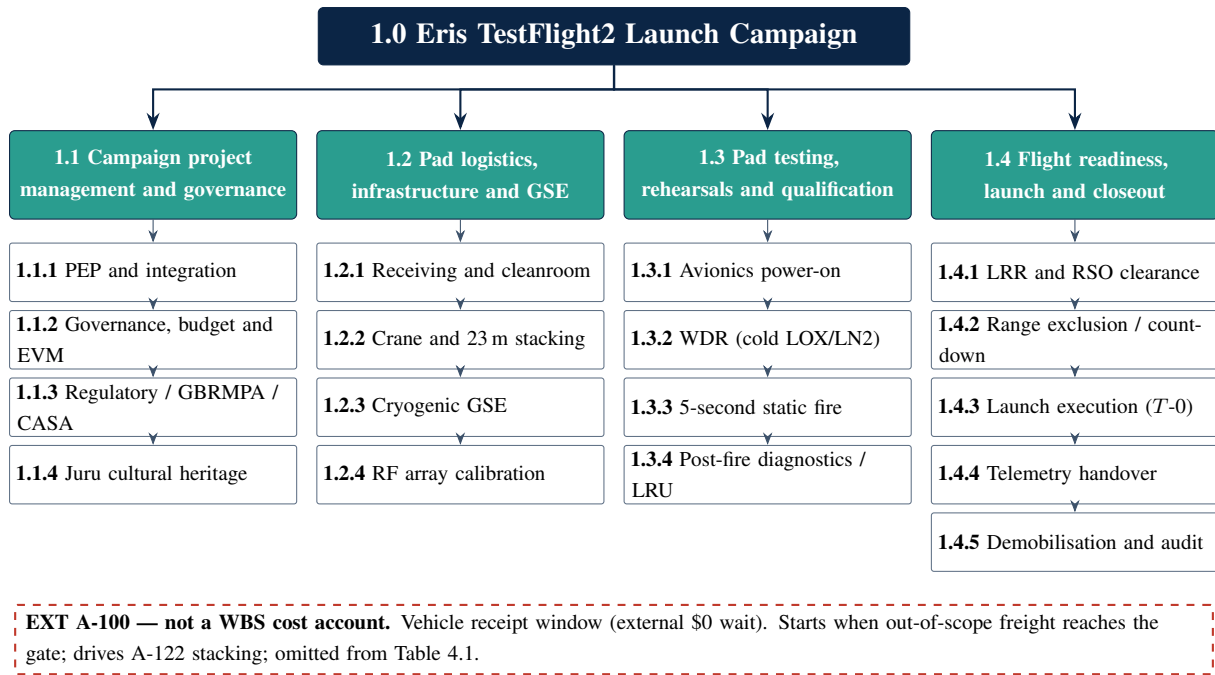


Figure 2.1: Work breakdown structure for Eris TestFlight2 launch-site operations. Seventeen Level-3 packages are the cost accounts in Table 4.1. A-100 is an external schedule bar, not a seventeenth-plus cost line.

### 3 Schedule management

The campaign schedule converts the WBS into a time-phased baseline for board approval. Dates are derived from a finish-to-start activity-on-node network on a seven-day pad calendar from 10 August to 1 December 2026. Approved  $T-0$  remains 15 November 2026. A five-day pull-in to 10 November is a contingent response to payload-window or airspace-slot risk; it is not the baseline.

#### 3.1 Network logic and float analysis

Precedence is modelled as activity-on-node with finish-to-start links and zero lag Project Management Institute (2021); Kerzner (2017). Textbook discrete-time notation writes  $EF = ES + D$  when  $EF$  is exclusive. On the Bowen calendar every displayed date is inclusive, so the implemented identities are

$$ES_i = \max_{j \in P_i}(EF_j) + 1, \quad EF_i = ES_i + D_i - 1, \quad (1)$$

$$LF_i = \min_{k \in S_i}(LS_k) - 1, \quad LS_i = LF_i - D_i + 1, \quad (2)$$

with  $P_i$  and  $S_i$  the predecessor and successor sets of activity  $i$ , and with  $ES$  equal to the campaign start where  $P_i$  is empty. Total float and free float then follow as

$$TF_i = LS_i - ES_i = LF_i - EF_i, \quad FF_i = \min_{k \in S_i} (ES_k) - EF_i - 1, \quad (3)$$

taking  $FF_i = TF_i$  when  $S_i$  is empty. Launch-critical means  $TF_i = 0$  on a path that sets  $T-0$  or closeout. Table 3.1 is the full pass and matches the locked register.

The forward pass produces two  $TF = 0$  chains that remain independent until hot fire. The vehicle path is A-100 (external gate receipt; freight out of scope), then A-122, A-123, A-132 and A-133. The permit path is A-113 into A-133. After A-133 the tail A-134–A-145 is a single critical chain. A-111 finishes on 2 October by an imposed Must-Finish-On (Gate 1) and is not launch-critical: the Week 4 charter already authorised mobilisation.

Path convergence is two staged merges. Wet Dress Rehearsal (A-132) cannot start until stacking (A-123), cryogenic GSE (A-124), avionics power-on (A-131) and Juru monitors (A-114) are complete—the technical merge. Static fire (A-133) then waits on A-132 *and* flight permits (A-113)—the regulatory merge. Launch Readiness Review (A-141) waits on post-fire diagnostics (A-134), not on A-133. Signing a flight-clearance gate on incomplete hot-fire data would break the hold-point logic required of an AS9100D system SAE International (2016).

Feeder float is not interchangeable. A-131 has one day into A-132 and is near-critical; A-124 has three. A-125 (RF) has 39 days and is parked in August–September so it does not sit on the October pad peak; moving it does not change 15 November. A-121 is the instructive case:  $TF = 3$  but  $FF = 0$ , so delay is absorbed before  $T-0$  only by consuming A-124's float, while A-124's early start moves immediately. Total float protects  $T-0$ ; free float protects the successor. Figure 3.1 draws the links. Crimson arrows are the  $TF = 0$  launch paths; the navy outline on A-111 is Gate 1.

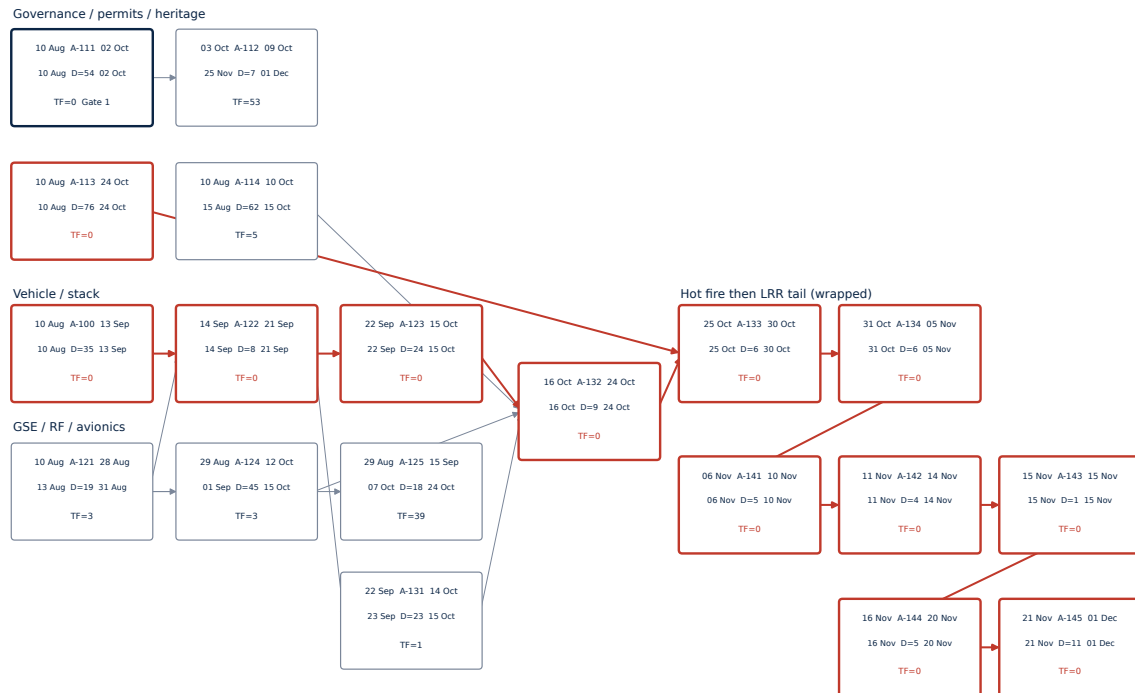


Figure 3.1: Activity-on-node network for the Bowen campaign. Node bands are  $ES$  /  $ID$  /  $EF$ ,  $LS$  / duration /  $LF$ , and  $TF$ . Crimson arrows are launch-critical finish-to-start links. The vehicle path and the A-113 permit path merge at A-133; A-141 waits on A-134, not on A-133. A-111 is Gate-1 constrained and is not launch-critical.

Table 3.1: Master campaign activity precedence and float register. Dates are inclusive.  $FF = \min(ES_{\text{succ}}) - EF - 1$  days, or  $FF = TF$  if there is no successor. A-111  $TF = 0$  is an imposed Gate 1 finish, not launch-critical.

| ID    | WBS    | Activity                        | $D$ | Pred.                      | ES     | EF     | LS     | LF     | TF | FF | CP   |
|-------|--------|---------------------------------|-----|----------------------------|--------|--------|--------|--------|----|----|------|
| A-100 | EXT    | Vehicle receipt window          | 35  | —                          | 10 Aug | 13 Sep | 10 Aug | 13 Sep | 0  | 0  | Y    |
| A-111 | 1.1.1  | PEP and integration baseline    | 54  | —                          | 10 Aug | 2 Oct  | 10 Aug | 2 Oct  | 0* | 0  | Gate |
| A-112 | 1.1.2  | EVM control stand-up            | 7   | A-111                      | 3 Oct  | 9 Oct  | 25 Nov | 1 Dec  | 53 | 53 | N    |
| A-113 | 1.1.3  | CASA / GBRMPA / airspace        | 76  | —                          | 10 Aug | 24 Oct | 10 Aug | 24 Oct | 0  | 0  | Y    |
| A-114 | 1.1.4  | Juru monitor mobilisation       | 62  | —                          | 10 Aug | 10 Oct | 15 Aug | 15 Oct | 5  | 5  | N    |
| A-121 | 1.2.1a | Site mobilisation and cleanroom | 19  | —                          | 10 Aug | 28 Aug | 13 Aug | 31 Aug | 3  | 0  | N    |
| A-122 | 1.2.1b | Receiving and unpacking         | 8   | A-100, A-121               | 14 Sep | 21 Sep | 14 Sep | 21 Sep | 0  | 0  | Y    |
| A-123 | 1.2.2  | Crane and vertical stacking     | 24  | A-122                      | 22 Sep | 15 Oct | 22 Sep | 15 Oct | 0  | 0  | Y    |
| A-124 | 1.2.3  | Cryo GSE piping and checkouts   | 45  | A-121                      | 29 Aug | 12 Oct | 1 Sep  | 15 Oct | 3  | 3  | N    |
| A-125 | 1.2.4  | RF telemetry calibration        | 18  | A-121                      | 29 Aug | 15 Sep | 7 Oct  | 24 Oct | 39 | 39 | N    |
| A-131 | 1.3.1  | Avionics power-on sweeps        | 23  | A-122                      | 22 Sep | 14 Oct | 23 Sep | 15 Oct | 1  | 1  | N    |
| A-132 | 1.3.2  | Wet Dress Rehearsal (cold)      | 9   | A-114, A-123, A-124, A-131 | 16 Oct | 24 Oct | 16 Oct | 24 Oct | 0  | 0  | Y    |
| A-133 | 1.3.3  | 5-second static fire            | 6   | A-132, A-113               | 25 Oct | 30 Oct | 25 Oct | 30 Oct | 0  | 0  | Y    |
| A-134 | 1.3.4  | Post-fire diagnostics and LRU   | 6   | A-133                      | 31 Oct | 5 Nov  | 31 Oct | 5 Nov  | 0  | 0  | Y    |
| A-141 | 1.4.1  | LRR and RSO clearance           | 5   | A-134                      | 6 Nov  | 10 Nov | 6 Nov  | 10 Nov | 0  | 0  | Y    |
| A-142 | 1.4.2  | Range exclusion and countdown   | 4   | A-141                      | 11 Nov | 14 Nov | 11 Nov | 14 Nov | 0  | 0  | Y    |
| A-143 | 1.4.3  | Launch execution ( $T-0$ )      | 1   | A-142                      | 15 Nov | 15 Nov | 15 Nov | 15 Nov | 0  | 0  | Y    |
| A-144 | 1.4.4  | Telemetry decrypt and handover  | 5   | A-143                      | 16 Nov | 20 Nov | 16 Nov | 20 Nov | 0  | 0  | Y    |
| A-145 | 1.4.5  | Demobilisation and closeout     | 11  | A-144                      | 21 Nov | 1 Dec  | 21 Nov | 1 Dec  | 0  | 0  | Y    |

### 3.2 Schedule baseline, Gantt visualisation and milestone integrity

The dates in Table 3.1 *are* the schedule baseline. Figure 3.2 plots the same  $ES/EF$  values, colours launch-critical bars crimson, and shows remaining total float as a teal whisker to  $LF$ . Permitting, heritage, mobilisation and GSE run through August and September; October is the physical peak (stack 15 October, WDR 16–24 October, static fire 25–30 October); November is the flight-clearance tail.

Two Week 4 collisions are closed here. The charter dated pad integration 15 October in Objective 1 and stacking as M-4 on 28 October; the PEP locks M-4 to A-123 on **15 October 2026**, restoring the cold-flow window before hot fire. Handover “within five days of lift-off” was also dated M-8 at 22 November (seven days after a 15 November  $T-0$ ); M-8 is now A-144 on **20 November**. Because the facility licence is already held, A-113 gates static fire, not stacking. Treating flight permits as a stacking predecessor was an initiation error.

Three gates sit on the baseline. Gate 1 is the PEP lock at A-111 on 2 October. Gate 2 is A-113 on 24 October: without those instruments A-133 cannot start, and that path is already  $TF = 0$ . Gate 3 is A-141 on 10 November. Crashing stacking without crashing A-113 therefore cannot pull  $T-0$ .

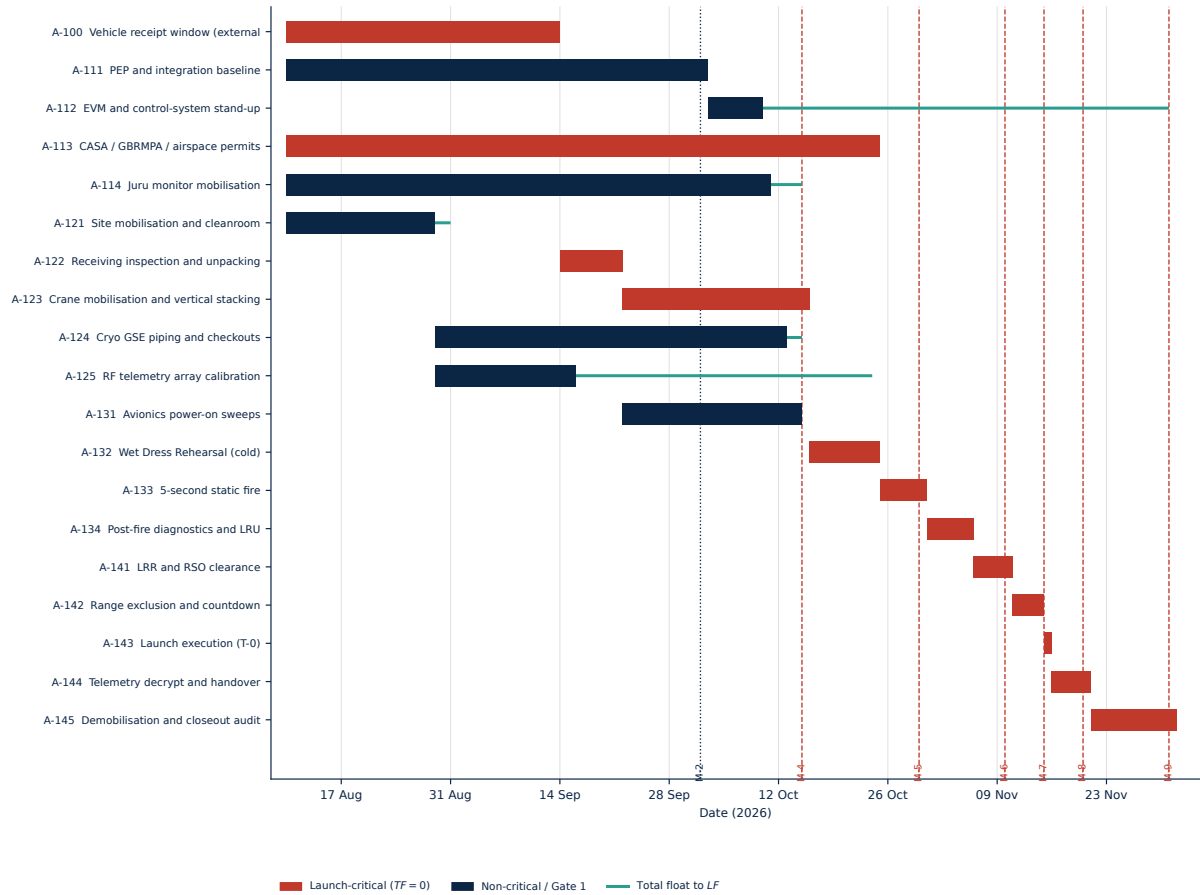


Figure 3.2: Baseline Gantt from 10 August to 1 December 2026, plotted from the same  $ES/EF$  values as Table 3.1. Crimson bars are launch-critical. Teal whiskers show total float to  $LF$ . Vertical markers are M-2 (navy, Gate 1) and M-4 through M-9. Option C is not drawn on this baseline.

### 3.3 Schedule compression: crashing and fast-tracking

Kerzner (2017) and Project Management Institute (2021) distinguish fast-tracking (overlap of sequential work, usually at zero direct cost) from crashing (added resource on critical work, paid as a cost slope). The Board question is not weather. It is whether  $T-0$  can move from 15 November to 10 November if R-12 is realised (payload LEO geometry valid only 10–12 November) or R-14 is realised (CASA NOTAM refuses the 15 November slot). Five days must come off a  $TF = 0$  path that actually sets lift-off. After the A-133 merge that path is the tail A-134–A-143. Crashing A-123 by five days, or overlapping A-123 with A-124, saves **zero** days on  $T-0$  while A-113 still finishes on 24 October. A-124 already runs in parallel with stacking.

The crash-cost slope on a crashed activity is

$$\text{Cost slope} = \frac{\Delta C}{\Delta T} = \frac{\text{crash cost} - \text{normal cost}}{\text{normal duration} - \text{crash duration}}. \quad (4)$$

Table 3.2 and Figure 3.3 compare three options that each claim five days. Only Option C is both

CPM-valid and acceptable against the LRR hold point.

Table 3.2: Schedule compression evaluation. All three options save five days on  $T-0$  if they work as drawn. Baseline  $T-0$  remains 15 November; Option C is a contingent draw on the \$225,000 reserve, not an addition to the \$1,500,000 base.

| Option                        | Activities modified                                                               | Days | Added cost      | Slope            | Secondary risks                                                               | Verdict    |
|-------------------------------|-----------------------------------------------------------------------------------|------|-----------------|------------------|-------------------------------------------------------------------------------|------------|
| A Pure fast-track             | A-134 → A-141 SS+1 (LRR overlaps diagnostics)                                     | 5    | \$0             | \$0/d            | LRR before diagnostic close-out; LRU rework; AS9100D hold-point failure       | Rejected   |
| B Pure crash                  | A-134 crashed<br>6 → 1 d                                                          | 5    | \$40,000        | \$8,000/d        | Fatigue; skipped thermal/electrical screens; recycle into A-141               | Rejected   |
| C Hybrid (if R-12/R-14 fires) | A-134<br>6 → 3 d (\$9,000);<br>A-141<br>5 → 3 d (\$12,000);<br>A-142<br>uncrashed | 5    | <b>\$21,000</b> | <b>\$4,200/d</b> | Staggered 10-hour shifts; residual diagnostic pressure, buffered by countdown | Contingent |

Option A buys five days at zero direct cost by starting A-141 one day after A-134 starts, so the review board would sit while diagnostics were still open. An LRU swap on day four either rubber-stamps incomplete evidence or recalls the review—a hold-point failure under AS9100D SAE International (2016). Option A is rejected.

Option B keeps finish-to-start logic but compresses A-134 from six days to one. The slope is  $\$40,000/5 = \$8,000$  per day. A 24-hour diagnostic war room on a vehicle that has just fired is a human-factors problem: thermal and electrical screens are the work most likely to be skipped. The slope is also the dearest of the three. Option B is rejected.

Option C splits the five days across two critical activities after the permit merge: three days off A-134 (\$3,000 per day, \$9,000) and two days off A-141 (\$6,000 per day, \$12,000). The weighted slope is  $\$21,000/5 = \$4,200$  per day, about half of Option B. Diagnostics remain three days rather than one; LRR remains a finish-to-start successor; A-142 is not crashed, so the four-day range and countdown buffer before  $T-0$  is kept. Fatigue is managed with staggered ten-hour shifts. If invoked, the dates become A-134 31 October–2 November, A-141 3–5 November, A-142 6–9 November, and A-143 on **10 November**. The \$21,000 is drawn from the \$225,000 contingency only if R-12 or R-14 is realised. It is not loaded into the \$1,500,000 base, and

Figure 3.2 is not redrawn.

Because Option C does not shorten A-132 or A-133, feeder floats do not tighten: A-131 stays at  $TF = 1$ , A-124 at  $TF = 3$ . No secondary critical path is created. Residual exposure is quality pressure on a shorter diagnostic window, which is why A-142 is left intact and why the option remains contingent.

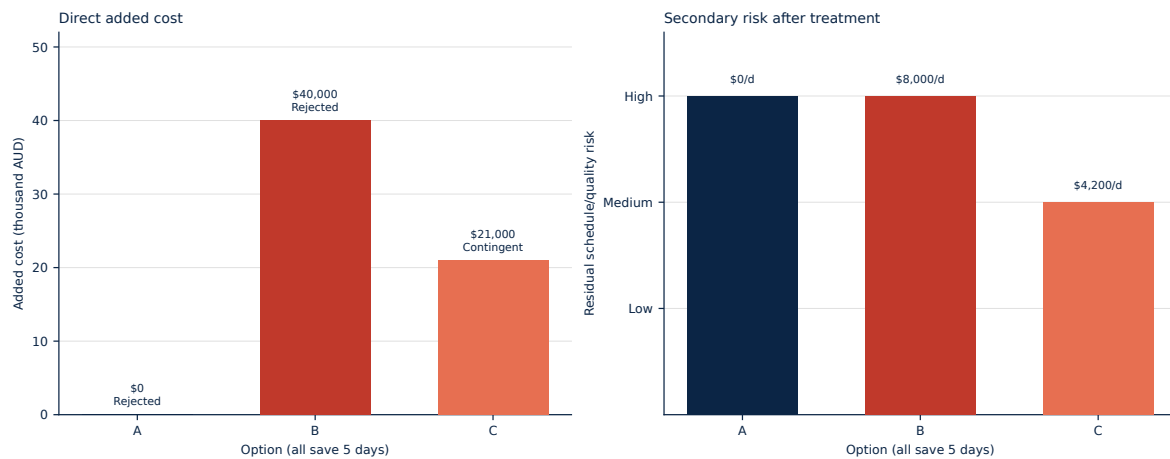


Figure 3.3: Compression trade-off for a five-day pull-in of  $T-0$ . Left: direct added cost. Right: residual schedule and quality risk after treatment. Option C is the only CPM-valid pull-in that keeps A-142 uncrashed and pays \$4,200 per day rather than \$8,000.

## 4 Cost management

The cost baseline is the WBS in Figure 2.1 priced at Level 3. Seventeen control accounts sum to a work estimate of \$1,500,000. Contingency of \$225,000 is sized from quantified risk, not from a round-number markup, so the authorised budget at completion is \$1,725,000.

### 4.1 Estimating basis and assumptions

The estimate is bottom-up at Level 3, with analogous plant and commodity rates beside the hours (Kerzner, 2017; Project Management Institute, 2021). Loaded 2026 Australian-dollar rates are used throughout: systems-engineering and project-management time at \$150/h, pad technicians at \$95/h, cryogenic and range-safety specialists at \$165/h, and surge technicians at \$100/h. Base pay follows the Manufacturing and Associated Industries Award with on-costs for superannuation, leave and workers compensation (Fair Work Ombudsman, 2020). The 100 t crawler for 23 m stacking is priced at \$4,200 per operating day, including operator, from published Queensland plant-hire schedules cross-checked against Rawlinsons (Rawlinsons, 2025).

Inclusions are the seventeen packages in Figure 2.1: mobilisation through demobilisation at Bowen. Factory manufacture, engine research and development, and freight to the receiving gate remain out of scope, so A-100 is a zero-dollar external wait rather than a cost account. The

four-person surge crew (\$22,400) is inside WBS 1.3.3 as planned capacity. The \$21,000 hybrid compression cost in Section 3.3 is a contingent draw against the \$225,000 reserve if R-12 or R-14 is realised; it is not in the \$1,500,000 performance measurement baseline.

A limited notice-to-proceed in the Week 4 charter authorised mobilisation from 10 August. Full cost and schedule lock at Gate 1 (A-111, 2 October). August–September field cost is planned spend, not unauthorised commitment.

## 4.2 WBS cost baseline

Table 4.1 is the rollup. Pad logistics (1.2) is \$518,000 (34.5% of the work estimate), pad testing (1.3) \$414,000 (27.6%), governance (1.1) \$300,000 (20.0%), and flight/closeout (1.4) \$268,000 (17.9%).

Table 4.1: Bottom-up cost baseline by Level-3 WBS package. Totals equal the work estimate of \$1,500,000. A-100 is omitted because it is not a cost account.

| WBS         | Package                      | Amount (AUD)     | Basis                                       |
|-------------|------------------------------|------------------|---------------------------------------------|
| 1.1.1       | PEP and integration          | 72,000           | 480 h PM @ \$150                            |
| 1.1.2       | Governance and EVM           | 48,000           | 320 h planner/cost @ \$150                  |
| 1.1.3       | CASA / GBRMPA permits        | 88,000           | Consultant + CASA cost-recovery             |
| 1.1.4       | Juru monitors and heritage   | 92,000           | Two monitors + survey                       |
| 1.2.1       | Receiving and cleanroom      | 68,000           | Technician hours + consumables              |
| 1.2.2       | Crane and 23 m stacking      | 218,000          | 24 d $\times$ \$4,200 = \$100,800 + riggers |
| 1.2.3       | Cryo GSE                     | 180,000          | Valves/piping + specialist crew             |
| 1.2.4       | RF array calibration         | 52,000           | Crew + analyser hire                        |
| 1.3.1       | Avionics power-on            | 58,000           | Engineers @ \$150/h + EGSE                  |
| 1.3.2       | WDR (cold)                   | 160,000          | Includes bulk LOX 25 t and LN2 \$68,500     |
| 1.3.3       | Static fire                  | 148,000          | Includes surge hire \$22,400                |
| 1.3.4       | Post-fire diagnostics        | 48,000           | Engineers + LRU handling                    |
| 1.4.1       | LRR and RSO                  | 38,000           | Review board                                |
| 1.4.2       | Range exclusion / marine     | 58,000           | Patrol boats + NOTAM                        |
| 1.4.3       | Launch and telemetry capture | 82,000           | Launch-day operations                       |
| 1.4.4       | Decrypt and handover         | 24,000           | Five-day engineering handover               |
| 1.4.5       | Demobilisation and audit     | 66,000           | Site restore + audit                        |
| <b>Base</b> |                              | <b>1,500,000</b> |                                             |

*Notes.* Labour: Award base plus on-costs (Fair Work Ombudsman, 2020). Crane and demobilisation: Rawlinsons and Queensland plant-hire (Rawlinsons, 2025). CASA line: agency cost-recovery (Civil Aviation Safety Authority, 2024). Surge \$22,400 =  $4 \times 7 \text{ d} \times 8 \text{ h} \times \$100$  and sits in 1.3.3, not in contingency.

The largest package is crane mobilisation and stacking (1.2.2, \$218,000) because A-123 occupies 24 calendar days on the launch-critical path: 24 days  $\times$  \$4,200 = \$100,800 of hire, with the balance in rigger labour. Wet Dress Rehearsal (1.3.2) carries \$68,500 of bulk liquid oxygen and nitrogen inside \$160,000. Static fire (1.3.3) carries the planned surge inside \$148,000.

Demobilisation (1.4.5) is a Rawlinsons-based site restore plus the closeout audit (Rawlinsons, 2025). Regulatory labour in 1.1.3 includes Civil Aviation Safety Authority cost-recovery (Civil Aviation Safety Authority, 2024). The seventeen lines sum to \$1,500,000 with no miscellaneous remainder. Package 1.2.1 remains one control account for A-121 and A-122.

### 4.3 Contingency

Contingency is not a 15% story that happens to look quantitative. The 15.00% ratio is the *result* of adding an expected-monetary-value (EMV) pool to a residual uncertainty allowance (Kerzner, 2017). Table 4.2 is the arithmetic. EMV uses the dollar-impact column of the risk register ( $EMV = P_{\%} \times I_{\$}$ ), not the 1–5 ordinal scores. Summing the eleven cost-bearing risks R-01, R-02, R-03, R-04, R-05, R-06, R-07, R-09, R-10, R-13 and R-14 gives \$175,500. An in-scope residual-uncertainty (RES) allowance of \$49,500 covers liquid-oxygen remote-delivery price variance, crane standby in wind, and unquantified pad anomalies. Vehicle freight is out of scope and is not in RES. The two components sum to \$225,000, which is 15.00% of \$1,500,000, and the authorised *BAC* is therefore \$1,725,000.

Table 4.2: Contingency build-up. The 15.00% ratio is the result of EMV plus an in-scope residual allowance, not an input markup.

| Component                                 | Amount (AUD)     | Rule                                              |
|-------------------------------------------|------------------|---------------------------------------------------|
| $\Sigma$ EMV of eleven cost-bearing risks | 175,500          | Dollar EMV from Sec. 6.2; not 1–5 scores          |
| RES uncertainty allowance                 | 49,500           | LOX price, crane wind, pad anomalies; not freight |
| <b>Contingency reserve</b>                | <b>225,000</b>   | 15.00% of base as a <i>result</i>                 |
| <b><i>BAC</i></b>                         | <b>1,725,000</b> | Base + contingency                                |

Several register rows are kept out of the \$175,500 so that money is not counted twice. R-08 (pad-civil defect) is contract-transferred under the design-and-construct package and carries no Gilmour EMV. R-11, R-15, R-16, R-17 and R-18 are schedule or HSE exposures without a residual dollar draw. R-12 (payload LEO window) carries a \$21,000 impact equal to Option C; its EMV is held at zero so the compression what-if is not also inside the pre-event total. If R-12 or R-14 is realised and Option C is invoked, the \$21,000 is drawn from the \$225,000 reserve. It is not added to the \$1,500,000 work baseline. There is no management reserve above *BAC*: unknown unknowns would require a board variation this plan does not authorise (Project Management Institute, 2021).

## 4.4 Time-phased budget and cash flow

The performance measurement baseline is the work estimate spread across the same dates as Table 3.1. Table 4.3 and Figure 4.1 are that spread. August and September (\$218,000 and \$284,000) are charter-authorised mobilisation. October is the peak month at \$657,000 because stacking completes (M-4, 15 October), Wet Dress Rehearsal runs, and the five-second static fire is executed. November carries range,  $T-0$  and most of demobilisation (\$315,000). December is a \$26,000 residual on 1 December, not a parking place for unused contingency.

Table 4.3: Time-phased planned value of the \$1,500,000 work baseline. Contingency is not monthly spend.

| Month                                    | Period PV (AUD) | Cumulative PV (AUD) |
|------------------------------------------|-----------------|---------------------|
| Aug 2026                                 | 218,000         | 218,000             |
| Sep 2026                                 | 284,000         | 502,000             |
| Oct 2026 (peak: stack, WDR, static fire) | 657,000         | 1,159,000           |
| Nov 2026                                 | 315,000         | 1,474,000           |
| Dec 2026                                 | 26,000          | <b>1,500,000</b>    |

The S-curve therefore terminates at \$1,500,000. Figure 4.1 draws  $BAC = \$1,725,000$  as a horizontal authorisation line. Planned value at the first physical gate (M-4, 15 October) is \$825,000—mid-October, between the September close of \$502,000 and the October close of \$1,159,000. That M-4 figure is the  $PV$  used in Section 7.5; it is not a fourteenth weekly data point invented after the fact. Option C's \$21,000 is excluded from every period in Table 4.3. If it is spent, it will appear as actual cost charged to contingency, not as a rewrite of the S-curve.

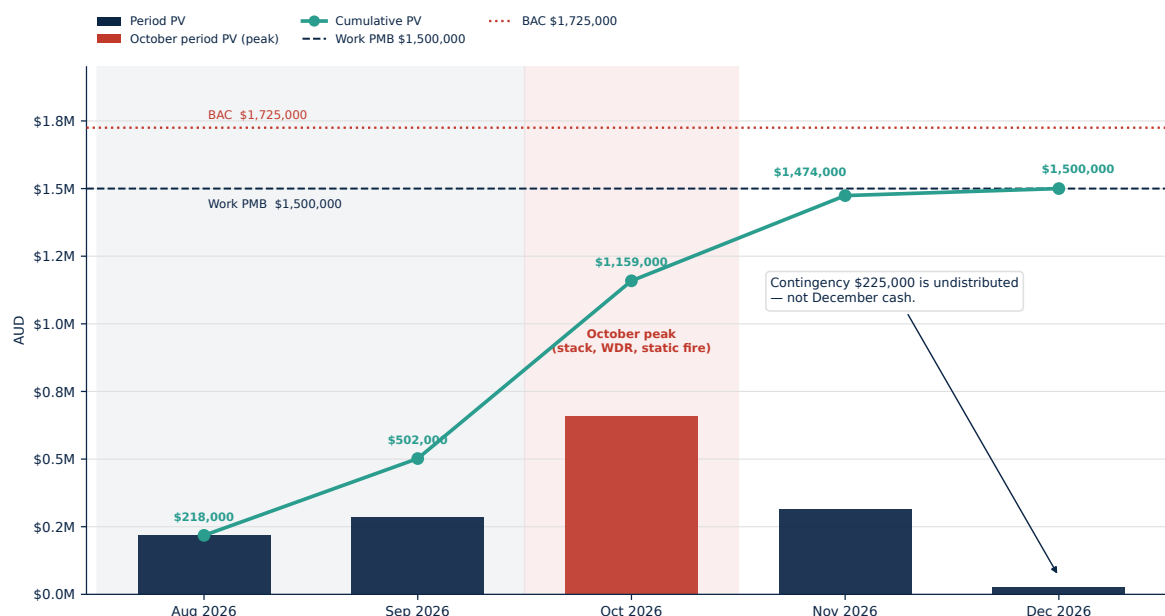


Figure 4.1: Cumulative planned-value S-curve for the Bowen campaign. Period bars are monthly PV; the teal line is cumulative PV and ends at the \$1,500,000 work baseline. The crimson dotted line is *BAC* (\$1,725,000) as an authorisation ceiling, not December cash. Contingency remains undistributed. Option C is not in the curve.

## 5 Resource management

Pad occupancy is a health-and-safety limit, not a roster convenience. The organic Bowen crew is eight certified technicians. The blast-zone, suit-air and access-control cap is twelve people. Eight is therefore the standing organisation; twelve is the occupancy ceiling that Figure 5.1 must never cross (Project Management Institute, 2021; Kerzner, 2017).

### 5.1 Roles and responsibilities

Accountability is split across two layers, with exactly one Accountable owner per Level-3 package in Table 5.1. The consultancy layer is Abdul Wahab Obeid (campaign PM and integration), Hadi Alkhub (planner and schedule), Ziyad Alahmari (risk, quality and HSE) and Ilsa Hashmi (cost, resources and the Juru commercial interface). The field layer is the Launch Director, Range Safety Officer, GSE Lead, Avionics Specialist, Juru Cultural Officer and the design-and-construct package manager. Consulted names are the people who must see the work product before the Accountable owner signs; they are not a second A.

The pattern matches the network. Governance packages sit with the consultancy: Abdul on the PEP and on CASA/GBRMPA permits, Ilsa on EVM, the Juru Cultural Officer on heritage with Ilsa consulted. Pad logistics sit with the people who physically control the lift and the interfaces: Hadi is Accountable for stacking because the dawn-lift window is a schedule control (R-06); GSE and D&C share cryogenic hardware; avionics owns RF and power-on. Hot oper-

ations (static fire, LRR,  $T-0$ ) are Accountable to the Launch Director with the RSO consulted, except range exclusion itself, which the RSO owns. Closeout audit is Ziyad's, so quality is not marking its own homework against a field lead.

Table 5.1: RACI for Level-3 packages. Exactly one Accountable owner per row. Field titles are as used on the pad; consultancy names are the Group 1 officers.

| WBS   | Package               | A               | R               | C                    |
|-------|-----------------------|-----------------|-----------------|----------------------|
| 1.1.1 | PEP and integration   | Abdul           | Hadi            | Ziyad, Ilsa          |
| 1.1.2 | Governance and EVM    | Ilsa            | Hadi            | Abdul                |
| 1.1.3 | CASA / GBRMPA permits | Abdul           | Hadi            | Launch Director      |
| 1.1.4 | Juru heritage         | Juru Officer    | Ilsa            | Abdul, Ziyad         |
| 1.2.1 | Receiving / cleanroom | GSE Lead        | D&C mgr         | Hadi, Ilsa           |
| 1.2.2 | Crane and stacking    | Hadi            | GSE Lead        | Launch Director, RSO |
| 1.2.3 | Cryo GSE              | D&C mgr         | GSE Lead        | Ziyad                |
| 1.2.4 | RF calibration        | Avionics        | Hadi            | Launch Director      |
| 1.3.1 | Avionics power-on     | Avionics        | GSE Lead        | Hadi                 |
| 1.3.2 | WDR (cold)            | GSE Lead        | Juru Officer    | Ziyad, LD, RSO       |
| 1.3.3 | Static fire           | Launch Director | GSE Lead        | RSO, Ziyad, Avionics |
| 1.3.4 | Post-fire / LRU       | Avionics        | GSE Lead        | Ziyad                |
| 1.4.1 | LRR and RSO           | Launch Director | RSO             | Abdul, Ziyad         |
| 1.4.2 | Range exclusion       | RSO             | Launch Director | Juru Officer         |
| 1.4.3 | Launch ( $T-0$ )      | Launch Director | RSO, Avionics   | Abdul                |
| 1.4.4 | Telemetry handover    | Avionics        | Hadi            | Ilsa                 |
| 1.4.5 | Demob and audit       | Ziyad           | D&C mgr         | Ilsa, Abdul          |

## 5.2 Capacity planning

Figure 5.1 is the weekly demand implied by Table 3.1 against those two headcount numbers. If RF calibration (A-125) remained stacked on Wet Dress Rehearsal and static fire, weeks 10–12 would peak at fourteen people: two above the HSE cap and six above the organic crew. That is R-13 in numeric form, not a generic “over-allocation.”

The baseline does two things, in that order. First, A-125 is placed in weeks 3–6 (29 August to 15 September) using its 39 days of total float. That move costs \$0 and does not change 15 November, because A-125 is not on the launch-critical path. Second, four surge technicians are hired for seven days inside WBS 1.3.3 at \$100/h loaded:  $4 \times 7 \times 8 \times \$100 = \$22,400$ , already inside the \$1,500,000 work estimate (Fair Work Ombudsman, 2020). The mitigated series then peaks at twelve in weeks 11–12, on the cap and never above it. Organic eight remains the standing crew; the extra four are a planned pulse, not evidence that the cap was quietly raised. Staggered ten-hour shifts in those two weeks keep twelve people from occupying the blast zone at once even though the daily roster is twelve. Option C, if invoked, shortens A-134 and

A-141 and does not redraw Figure 5.1: the surge technicians remain a 1.3.3 base cost, not a compression cost. Moving A-121 or A-124 would consume feeder float without lowering the week-11 peak, so they are not used as capacity tools.



Figure 5.1: Weekly pad demand versus capacity. Hashed crimson bars are the unmitigated series (peak 14 in weeks 10–12). Teal bars are the mitigated baseline (peak 12) after moving A-125 into weeks 3–6 and adding four surge technicians in 1.3.3. The navy dashed line is the organic crew of eight; the crimson dotted line is the HSE occupancy cap of twelve. Eight is not a hard pad limit.

### 5.3 Conflict resolution and fallback

Resource conflicts escalate on the work, not on personalities. Tier 1 field leads have four hours to level, stagger or borrow from float without spending money. Tier 2 is the campaign PM: twelve hours to choose overtime, hire or resequence, with the cost consequence going to Ilsa so that Table 4.1 stays the book of record. Tier 3 is the Executive Board, with a 24-hour SLA, for any decision that consumes contingency or moves M-5 (30 October) or M-7 (15 November). Those gates are the same dates as Section 3; they are not a second calendar.

Team-member fallback is separate and is the A1 defect closed here. If a member is non-responsive for more than 48 hours, the Lead Planner audits the artefacts within 12 hours, the PM issues a WBS-reassignment addendum, and the Buddycheck PAF is locked against Teams and version history so that contribution evidence cannot be rewritten after the fact. That procedure reallocates packages in Table 5.1; it does not authorise a silent change to *T-0* or to *BAC*.

## **6 Risk management**

### **6.1 Risk approach and scales**

### **6.2 Integrated risk register**

### **6.3 Top risks, contingency, float and delivery model**

## **7 Delivery and control**

### **7.1 Governance and reporting**

### **7.2 Delivery model and procurement**

### **7.3 Quality**

### **7.4 Health, safety and environment**

### **7.5 Earned value management**

## **8 Integration and professionalism**

### **8.1 Document consistency**

## 8.2 Presentation and referencing

### References

### References

Kerzner, H. (2017). *Project management: A systems approach to planning, scheduling, and controlling* (12th ed.). John Wiley & Sons.

Project Management Institute. (2021). *A guide to the project management body of knowledge (PMBOK® guide)* (7th ed.) and *The standard for project management*. Project Management Institute.

SAE International. (2016). *Quality management systems—Requirements for aviation, space, and defense organizations (AS9100D)*. SAE International.

Rawlinsons. (2025). *Rawlinsons Australian construction handbook*. Rawlinsons Publishing.

Fair Work Ombudsman. (2020). *Manufacturing and Associated Industries and Occupations Award 2020 (MA000010)*. Fair Work Commission.

Civil Aviation Safety Authority. (2024). *Cost recovery implementation statement: Aviation safety regulatory services*. Civil Aviation Safety Authority.

## A Contribution appendix

| Team member                  | Primary deliverables                               | %            |
|------------------------------|----------------------------------------------------|--------------|
| Abdul Wahab Obeid (s5303839) | Executive summary; integration; final QA           | 25.0         |
| Hadi Alkhub (s5331680)       | Scope; schedule; network logic; Gantt; compression | 25.0         |
| Ziyad Alahmari (s5375826)    | Risk register; quality and HSE                     | 25.0         |
| Ilsa Hashmi (s5310047)       | Cost; resources; capacity planning                 | 25.0         |
| <b>Total</b>                 |                                                    | <b>100.0</b> |

We confirm that the contribution percentages above are agreed by all members.

Abdul Wahab Obeid

Signature: \_\_\_\_\_

Date: \_\_\_\_\_

Ziyad Alahmari

Signature: \_\_\_\_\_

Date: \_\_\_\_\_

Hadi Alkhub

Signature: \_\_\_\_\_

Date: \_\_\_\_\_

Ilsa Hashmi

Signature: \_\_\_\_\_

Date: \_\_\_\_\_